Railway Ticket Booking System

Aim :

To analyze the Software Development Life Cycle (SDLC) models used in the development of a Railway Ticket Booking System and identify the most suitable SDLC model for its successful implementation.

Objective:

- Study different SDLC models.
- Analyze the development process of a Railway Ticket Booking System.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

Selected Software Application:

(Railway Ticket Booking System)

- A Railway Ticket Booking System is an online software application that enables passengers to search trains, check seat availability, book tickets, make secure online payments, cancel bookings, and track booking status. It provides a fast, reliable, and user-friendly platform for railway passengers while reducing manual work at reservation counters.
- The system supports multiple users simultaneously, maintains passenger records, generates e-tickets, calculates fares automatically, and provides real-time train schedules. It improves operational efficiency and enhances the travel experience by offering convenient digital services.

Functional Requirements:

- User Registration
- Secure Login
- Search Trains
- Check Seat Availability
- Select Class and Journey Date
- Book Tickets
- Online Payment
- Download E-Ticket
- Cancel Ticket
- View Booking History

- Track PNR Status

Railway Staff Module:

- Manage Train Schedule
- Update Seat Availability
- Update Train Status
- Verify Passenger Details
- Generate Reservation Reports

Admin Module:

- User Management
- Train Management
- Route Management
- Fare Management
- Report Generation
- Analytics Dashboard
- System Monitoring

Why Agile is Best?

Reasons

- Passenger requirements change frequently.
- New booking features are added regularly.
- Train schedules are updated continuously.
- Bug fixes need quick deployment.
- Mobile applications require frequent updates.
- Continuous customer feedback improves system performance.
- Agile enables faster software delivery with iterative development.
- Regular testing improves software quality and reliability.

Agile Development Process:

Requirement Gathering

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Sprint Planning

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System Design

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Development

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Testing

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Deployment

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Customer Feedback

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Next Sprint

Results

- The Railway Ticket Booking System requires continuous feature updates and high system availability.
- Agile SDLC supports iterative development, faster releases, and continuous user feedback.
- DevOps complements Agile through continuous integration, automated testing, and deployment.
- The combination of Agile and DevOps ensures secure ticket booking, faster software delivery, improved reliability, and better passenger satisfaction.

Conclusion

- The case study concludes that selecting the appropriate SDLC model depends on the project requirements. Since a Railway Ticket Booking System requires continuous updates, secure online transactions, real-time train information, and frequent customer interaction, the Agile SDLC model is the most suitable approach. When integrated with DevOps practices, it enables continuous integration, automated deployment, rapid bug fixing, enhanced security, and improved software quality. This combination ensures a reliable, scalable, and efficient railway reservation system that meets the evolving needs of passengers and railway authorities.